

PREDICTING DEFECTS IN 3D PRINTED PARTS

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Problem statement

The goal is to predict whether a 3D printed part will be defective based on various factors like nozzle temperature, print speed, material type, humidity, and printer type. Using machine learning, we aim to build a model that classifies prints as defective or not defective (binary classification). Key objectives include preprocessing data, training multiple classification models (like Random Forest or XGBoost), evaluating performance, and deriving insights for improving print quality and reducing waste.

Features

1. Layer Thickness

Affects surface finish and strength; thicker layers print faster but may reduce precision and cause defects.

2. Nozzle Temperature

Determines how well the material melts and bonds; incorrect temps can cause clogging or weak layers.

3. Bed Temperature

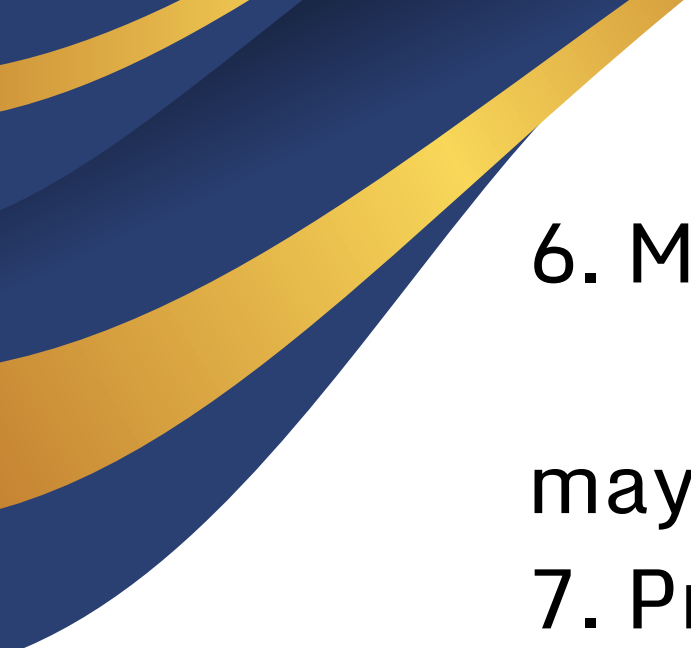
Helps the print stick to the bed; poor adhesion can lead to warping or failed prints.

4. Print Speed

Faster speeds reduce print time but can compromise quality and increase error rates.

5. Material Type

Different materials have unique melting points and behaviors; some are more prone to defects.



6. Model Complexity:

Complex models with fine details are harder to print accurately and may have higher defect risk.

7. Print Time:

Longer print times increase exposure to environmental changes and printer errors.

8. Printer Type:

Different printers have varying precision, stability, and reliability, affecting defect rates.

9. Environment Humidity:

Moisture in the air can affect material properties, especially hygroscopic filaments like nylon.

10. Cooling Method:

Cooling impacts how fast material solidifies; poor cooling can cause warping or weak structures.



Algorithms

Logistic Regression

K-Nearest Neighbors (KNN)

Decision Tree Classifier

Random Forest Classifier

Gradient Boosting Classifiers

Support Vector Machine (SVM)



Best algorithms

Linear regression

Predicting benefits :

Cost Reduction: Minimize material waste and printing time by identifying potential defects before they occur.

Improved Quality: Optimize printing parameters to consistently produce high-quality parts.

Efficiency: Automate quality control processes, saving time and labor costs.

Reduced Downtime: Prevent printer malfunctions and defective prints, leading to more continuous production.

Data-Driven Insights: Gain actionable insights into the factors affecting print quality for better decision-making.

Competitive Advantage: Offer more reliable, defect-free products, enhancing customer satisfaction and trust.

Questions & Answers

1. Question: How does nozzle temperature influence defect prediction in 3D printed parts?

Answer: Nozzle temperature affects the melting and flow of the printing material. Incorrect temperatures can cause poor layer adhesion or material burning, which increases the likelihood of defects. Hence, it's a key feature in defect classification models.

2. Question: Why is material type important in predicting defects in 3D printing?

Answer: Different materials have unique thermal and mechanical properties, which influence print quality. Some materials may warp or shrink more, making them prone to defects under certain settings. The model learns these patterns to predict outcomes accurately.

The background features a light gray field with a series of thin, white, wavy lines that create a sense of movement. In the corners, there are decorative elements: the top-left and bottom-right corners have thick, flowing bands of dark blue and gold, while the bottom-left corner contains a faint, white grid pattern.

THANK YOU